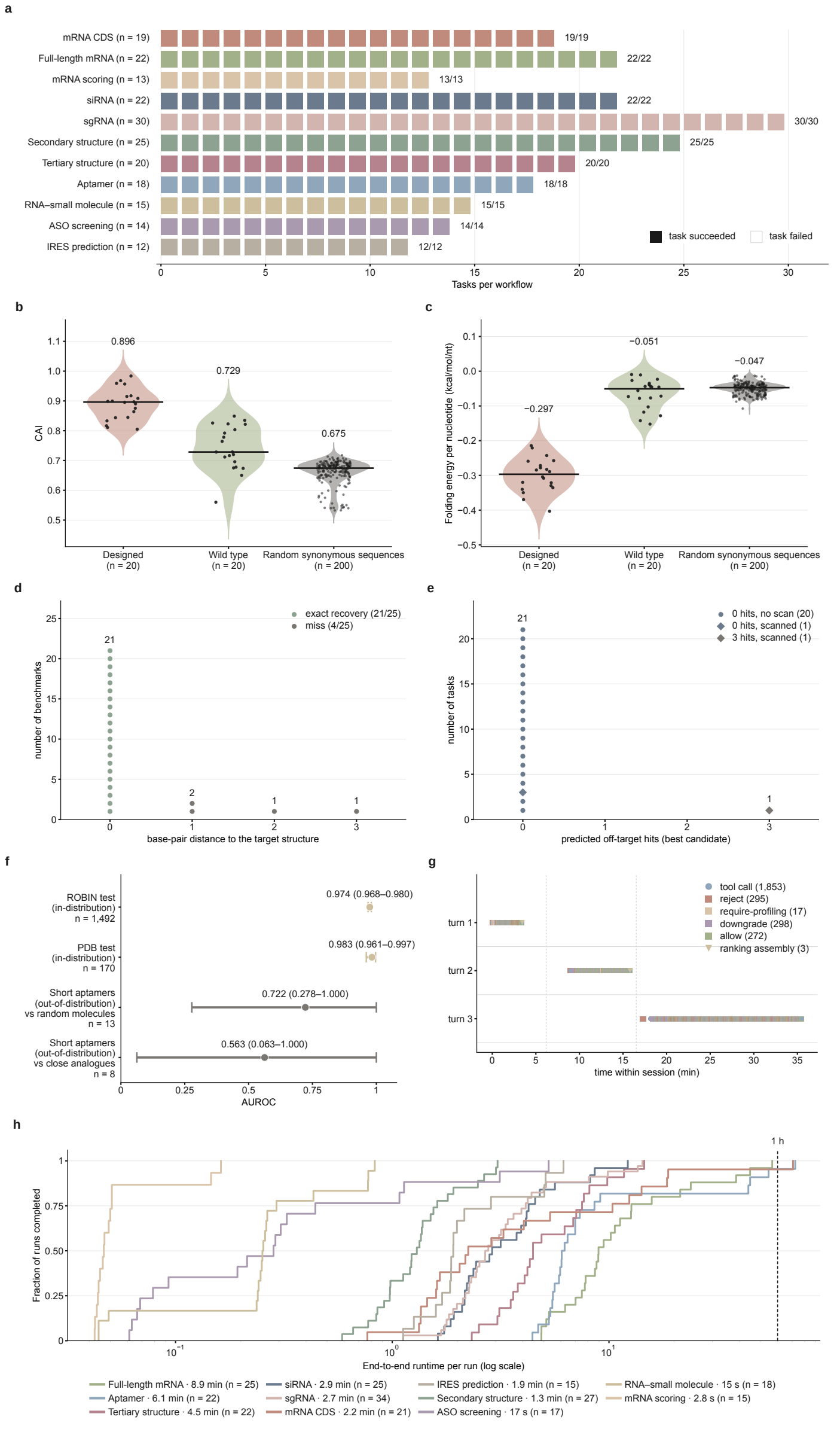

a
mRNA CDS (n = 19)
Full-length mRNA (n = 22)
mRNA scoring (n = 13)
siRNA (n = 22)
sgRNA (n = 30)
Secondary structure (n = 25)
Tertiary structure (n = 20)
Aptamer (n = 18)
RNA–small molecule (n = 15)
ASO screening (n = 14)
IRES prediction (n = 12)
19/19
22/22
13/13
22/22
30/30
25/25
20/20
18/18
15/15
14/14
12/12
task succeeded
task failed
Tasks per workflow
b
CAI
0.896
0.729
0.675
Designed (n = 20)
Wild type (n = 20)
Random synonymous sequences (n = 200)
c
Folding energy per nucleotide (kcal/mol/nt)
−0.297
−0.051
−0.047
Designed (n = 20)
Wild type (n = 20)
Random synonymous sequences (n = 200)
d
exact recovery (21/25)
miss (4/25)
number of benchmarks
base-pair distance to the target structure
e
0 hits, no scan (20)
0 hits, scanned (1)
3 hits, scanned (1)
number of tasks
predicted off-target hits (best candidate)
f
ROBIN test (in-distribution) n = 1,492
0.974 (0.968–0.980)
PDB test (in-distribution) n = 170
0.983 (0.961–0.997)
Short aptamers (out-of-distribution) vs random molecules n = 13
0.722 (0.278–1.000)
Short aptamers (out-of-distribution) vs close analogues n = 8
0.563 (0.063–1.000)
AUROC
g
tool call (1,853)
reject (295)
require-profiling (17)
downgrade (298)
allow (272)
ranking assembly (3)
turn 1
turn 2
turn 3
time within session (min)
h
1 h
Fraction of runs completed
End-to-end runtime per run (log scale)
Full-length mRNA · 8.9 min (n = 25)
siRNA · 2.9 min (n = 25)
IRES prediction · 1.9 min (n = 15)
RNA–small molecule · 15 s (n = 18)
Aptamer · 6.1 min (n = 22)
sgRNA · 2.7 min (n = 34)
Secondary structure · 1.3 min (n = 27)
mRNA scoring · 2.8 s (n = 15)
Tertiary structure · 4.5 min (n = 22)
mRNA CDS · 2.2 min (n = 21)
ASO screening · 17 s (n = 17)